



Can Large Language Models Develop High-Stakes Physics Exam Items? A Comprehensive Study of Cognitive and Psychometric Efficacy

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Received: 29 April 2025 / Accepted: 18 December 2025
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Abstract

High-stakes assessment is crucial for evaluating student performance and making significant educational decisions. Traditionally, the development of test items for such examinations has relied on manual development by subject matter experts. However, Automated Item Generation (AIG) using Large Language Models (LLMs) has emerged as a promising alternative, though systematic research on their application in high-stakes assessments, particularly in STEM fields like physics, is limited. High-stakes physics assessments must evaluate a range of cognitive skills, from basic recall to advanced analytical thinking. Previous AIG studies have predominantly focused on lower-order cognitive skills, neglecting higher-order thinking. Moreover, the psychometric quality of items generated by LLMs has not been thoroughly validated, raising concerns about their validity and reliability for high-stakes contexts. To explore this, we investigated LLM capabilities in generating physics test items suitable for Nigeria's Unified Tertiary Matriculation Examination (UTME). Based on our preliminary findings, the current study utilized Gemini 2.0 flash and instructional prompting technique for item generation. Bloom's taxonomy was used as a framework to generate items of different cognitive levels in line with the UTME blueprint. The quality of the generated items was evaluated through expert reviews using a six-criteria rubric and pilot testing with 527 final-year high school students. Psychometric analysis, based on Classical Test Theory and Item Response Theory, confirmed the high quality of the LLM-generated physics test, demonstrating unidimensionality and good psychometric properties. This study demonstrates that under controlled conditions and with expert review, LLMs can produce high-stakes physics MCQ items whose psychometric qualities (difficulty, discrimination) are comparable to those of a real UTME physics test. This suggests LLMs have potential as cost-effective item-generation tools, for educational institutions and testing organizations.

Keywords High-stakes assessment · Automated item generation (AIG) · Large language models (LLMs) · Psychometric analysis · Bloom's taxonomy · Item response theory

Introduction

High-stakes assessment is a crucial tool in educational assessment, used to evaluate student performance, make decisions, measure strengths, determine placement, and assess readiness for further education or professional growth. It aims to establish standardized criteria for equitable access to opportunities and foster accountability

within educational systems (Heilig & Nichols, 2013). In STEM (Science, Technology, Engineering, and Mathematics) subjects, such as physics, this assessment is particularly challenging due to the need to evaluate both theoretical understanding and practical problem-solving skills (Bao & Koenig, 2019). This issue is particularly prevalent when testing the validity, reliability, and fairness of test items. For instance, the Unified Tertiary Matriculation Examination (UTME) in Nigeria, administered by the Joint Admissions and Matriculation Board (JAMB), is a high-stakes examination that determines admission to tertiary institutions for millions of candidates annually. The physics component of the UTME is especially demanding, as it requires students to demonstrate mastery of a broad range of complex topics, often leading to poor performance among candidates (Ijah

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et al., 2025). Therefore, items must be crafted efficiently to discriminate among applicants with different abilities, providing a reliable basis for the competitive selection process that characterizes university admissions in Nigeria.

Traditionally, the development of high-quality test items for high-stakes examinations has been a labor-intensive and time-consuming process, relying heavily on manual item generation by subject matter experts. This approach faces significant challenges, including the limited pool of expert item writers, inconsistencies across exam versions, and difficulties in adapting to evolving curricula (Bennett, 2011). These limitations underscore the need for innovative ways to enhance the efficiency, scalability, and quality of item generation for high-stakes examinations.

Automated Item Generation (AIG) has emerged as a promising solution to these challenges, leveraging advances in artificial intelligence (AI) and natural language processing (NLP) to automatically create test items. AIG offers several advantages, including reduced development time and costs, improved scalability, and the ability to generate large pools of diverse, high-quality items (Falcão et al., 2023). Recent advancements in Large Language Models (LLMs), such as GPT-3.5 and GPT-4, have further expanded the potential of AIG by enabling the generation of context-based, cognitively diverse test items (Bulut et al., 2024; Mead & Zhou, 2024).

In STEM disciplines, problem-solving exercises have long been central to their assessment design. Studies show that such items engage students in quantitative problem solving and thereby significantly enhance their conceptual understanding by requiring them to actively apply equations and draw inferences (Chan et al., 2025; Küchemann et al., 2023; Bhandari et al., 2024). Hence, recognizing the critical role of problem-solving tasks in these fields, substantial efforts have been directed toward expanding physics test item banks using AIG techniques. This procedure creates room for automating item generation methods to produce a diverse array of assessment items that align with learning objectives.

However, despite these advancements, there remains a lack of systematic research on the application of LLMs in generating items for high-stakes examinations, particularly in STEM subjects like physics. The design and delivery of high-quality test items for high-stakes examinations, such as the UTME, present significant challenges, especially in physics, where assessments evaluate a wide range of cognitive skills. LLM-based AIG has shown promise in addressing these challenges, but existing approaches have primarily focused on lower-order cognitive skills (Mitani, 2021). Also, the psychometric quality of AIG-generated items remains unvalidated, raising questions about their validity, reliability, and suitability for high-stakes assessments. Additionally,

there is a lack of comprehensive validation frameworks for evaluating the quality of AIG-generated items.

This study aims to address these gaps by exploring the use of LLMs in generating physics test items for the UTME, focusing on their alignment with Bloom's taxonomy, evaluating their psychometric quality, and ascertaining their reliability. Two questions guided the study:

1. What are the psychometric qualities (difficulty, discrimination, distractor functioning) of the LLM-generated physics items, and how do they compare to high-stakes exam standards?
2. To what extent can LLMs reliably generate high-quality physics items suitable for UTME examinations, as evaluated by subject matter experts and through pilot testing?

Literature Review

Automatic Item Generation in Educational Assessment

Automated Item Generation (AIG) has also gained ground in the contexts of large-scale, high-stakes assessment, which require many items with high psychometric properties. Traditional AIG approaches have a long-standing history in educational measurement and psychometrics, predating the advent of LLMs. These methods have relied heavily on template-based systems wherein subject matter experts construct detailed item models that include fixed and variable components and then create several items or a family of items from the main item (Kurdi et al., 2020; Someshwar, 2024). This approach, often classified as weak AIG, allows for rapid item production while maintaining a consistent structure, but it also results in items that are highly similar and, at times, psychometrically interdependent (Arendasy et al., 2012). On the other side, strong AIG methods attempt to incorporate cognitive theories and models for the item generation process (Embretson, 2006b). Although strong AIG can potentially produce items with more desirable and targeted psychometric properties, it requires substantial theoretical grounding and computational resources. So, despite the advantages of AIG, these methods are not without limitations. While traditional AIG can efficiently produce a large volume of items, the process often requires extensive human review to validate content accuracy, ensure alignment with cognitive objectives, and eliminate inadvertent violations of fundamental item-writing principles (Gierl et al., 2021; Chan et al., 2025).

LLMs have revolutionized AIG by enabling the creation of multiple-choice items using prompts without the need

for explicit template creation. For instance, Küchemann et al. (2024) examined the reliability and validity of ChatGPT 3.5 generated kinematics concept items in physics by comparing them directly to the Force Concept Inventory. They found that uncontrolled, fully automated prompts can yield medium-difficulty items with strong convergent validity, but their work is confined to a single topic (kinematics).

While the integration of LLMs into AIG presents an opportunity to enhance item generation efficiency and expand content diversity, current implementations are constrained by issues related to psychometric quality, difficulty calibration, domain relevance, and factual accuracy (Mead, 2024). Most importantly, to date, it is not clear if LLMs possess an intrinsic mechanism to adjust item difficulty, like experienced human item writers. Also, there is much study needed to focus on refining prompt engineering techniques that allow the implementation of the full algorithm of generating items for high-stakes examination. Therefore, there are still limited understanding of how well LLMs especially the recent iterations such as Gemini 2.0 flash can generate items when prompted to align with educational taxonomies, such as the widely used Bloom's taxonomy for cognitive tasks in the context of high-stakes assessments.

Bloom's Taxonomy and Cognitive Levels in AIG

Bloom's taxonomy serves as a foundational framework to address the need for diverse cognitive demands in high-stakes assessments. Developed by Bloom (1964) and later revised by Krathwohl and Anderson (2010), the taxonomy categorizes learning objectives into six hierarchical levels that range from lower-order cognitive skills, such as remembering and understanding, to higher-order skills, such as applying, analyzing, evaluating, and creating. By structuring test items around Bloom's taxonomy, educators can ensure that assessments not only test factual knowledge but also include higher cognitive engagement (Krathwohl & Anderson, 2010; García-Ramírez, 2024). Higher-order cognitive levels test students' abilities to analyze experimental data, assess scientific reasoning, and apply theoretical concepts to interpret real-world situations (Sagala & Andriani, 2019). Such abilities are crucial for students pursuing careers in STEM.

There are various approaches to categorizing items according to cognitive levels, such as statistical methods utilizing natural language processing (NLP) techniques or machine learning (ML) (Abduljabbar & Omar, 2015). A recent study applied LLM for item classification (Doughty et al., 2024). Another approach is manual classification using subject matter experts (SMEs), which involves matching the keywords in the item with the cognitive verbs associated with each level of the taxonomy and determining the cognitive level (Mohammed & Omar, 2020).

The current LLM-based AIG approach has not been extensively studied for its integration of educational taxonomy in item generation or its ability to generate diverse items of cognitive abilities and levels for high-stakes assessment. A preliminary study by Omojekunola and Kardanova (2024) attempted to address this gap by incorporating Bloom's taxonomy into the LLM-based AIG pipeline, but it was limited in scope and focused on only two cognitive levels. A notable study by Doughty et al. (2024) explored the use of LLMs for generating multiple-choice questions (MCQs) in programming education, using Bloom's taxonomy to classify learning objectives into specific cognitive levels. However, this study did not apply Bloom's taxonomy to categorize the items or investigate whether LLMs could produce items that span a diverse range of cognitive abilities. The current study builds on these findings by focusing on all cognitive levels relevant to physics at the MCQ level, including remembering, understanding, applying, analyzing, and evaluating. The study aims to create a more comprehensive and cognitively diverse assessment tool for high-stakes examinations like the UTME, effectively measuring a wide range of cognitive skills.

Items Evaluation by Subject Matter Experts

Expert evaluation plays a critical role in ensuring the quality, validity, and reliability of test items across various educational and assessment contexts. Subject matter experts (SMEs) are often employed to review and refine test items, as their domain-specific knowledge and experience enable them to assess the clarity, correctness, and alignment of items with educational objectives (Haladyna & Rodriguez, 2013; Luecht, 2012). Also, despite the advances in LLM, there is a considerable concern among researchers on fairness and bias (gender, race, culture, etc.). LLMs inherit biases from training data, raising fairness concerns in testing. Deploying AIG in high-stakes contexts raises important regulatory and ethical challenges, hence, the need for item evaluation. Expert evaluation is particularly important in high-stakes assessments, where the consequences of flawed items can significantly impact students' academic and career trajectories.

The current LLM based AIG presents the dire need for item evaluation to assess the quality of LLM-generated items by SMEs. Structured frameworks like rubrics and checklists have been widely adopted to standardize expert evaluations (Küchemann et al., 2023; Doughty et al., 2024). For instance, Küchemann et al. (2023) employed a six-criteria framework to evaluate items in physics generated by prospective teachers with ChatGPT 3.5 and traditional textbook items using predefined criteria such as specificity, clarity, correctness, context, relevance, and overall quality.

Therefore, building on Chan et al. (2024), Küchemann et al. (2023) and Küchemann et al. (2024), this study adopts a six-criteria rubric. Specifically, this study extends the rubric of Küchemann by alignment with Bloom's taxonomy, which is critical for high-stakes assessment. Unlike Doughty et al. (2024), who included students as raters and thus risked introducing bias, this study engaged independent SMEs from Nigeria.

Piloting and Psychometric Analysis in AIG

In addition to the subject's expert evaluation, which provides qualitative information about the test items, psychometric analysis is another method to quantitatively assess the quality of AIG items. Hence, piloting and psychometric analysis constitute critical phase in the validation of test items. It provides empirical evidence of item quality and alignment with targeted cognitive constructs. Most importantly, the process ensures that the test items function effectively and accurately measure their intended purpose (Streiner et al., 2024). Despite their importance, only a few studies have conducted piloting and psychometric analysis of AIG items.

The two common approaches for analyzing test items are the Classical Test Theory (CTT) Framework and the Item Response Theory Framework (IRT). These procedures provide empirical insights into item difficulty, discrimination, distractor functionality, and test reliability, which are critical for constructing valid assessments. Although CTT often suffers from limitations such as a high dependency on samples and lacks some depth of psychometric analysis (Kyllonen et al., 2024), it can still be used effectively. For example, a recent study by Falcão et al. (2023) demonstrated that AIG items meet the rigorous standards required for high-stakes testing and are of similar or even higher quality compared to traditionally written items. However, there remains some inconsistency in the literature concerning the quality of distractors. While some studies indicate AIG-generated distractors may be of slightly lower quality, others report no significant differences (Pugh et al., 2020). The discrepancy may be due to improvements in AIG techniques. For IRT approach, example of such study is the study of Bhandari et al. (2024) in mathematics conducted to compare textbook items with LLM items. The study evaluated the ability of LLM items to differentiate between various ability levels and assessed their discriminating power. The results indicated that LLM items were comparable to human-authored textbook items in assessing students' abilities in college algebra.

To date, to the best of our knowledge on available studies on the LLM-based AIG approach, no study has been conducted on the piloting and psychometric testing of physics items generated by LLMs. This gap in the literature

highlights a critical need for research that not only develops these items but also comprehensively evaluates their quality and effectiveness. Given the increasing use of LLMs in assessment, it is essential to ensure that such items meet established psychometric standards and perform reliably in diverse testing environments.

Methodology

This section outlines the methodology for evaluating the effectiveness of LLMs in generating physics test items for UTME high-stakes assessments.

The Unified Tertiary Matriculation Examination (UTME) is Nigeria's high-stakes university-admissions test administered annually by the Joint Admissions and Matriculation Board (JAMB), to determine eligibility for undergraduate entry into Nigerian universities, polytechnics, and colleges of education. In 2025, about 2 million candidates registered for the test (Adewale 2025). The current UTME format 60 items in Use of English and 40 items for each of the three other chosen subjects (Physics, mathematics, Accounting etc.) administered within an approximate two-hour session, within the broader examination structure, and candidates must meet minimum entry requirements and be of eligible age (16 and above) as specified by the Board.

Given the large number of candidates that sit for the UTME each year and that Physics performance materially influences entry into competitive STEM programmes, any automated item-generation pipeline intended for use in this context must guarantee strict blueprint alignment, comprehensive expert review, and thorough psychometric calibration prior to operational deployment. Consequently, the domain balance, cognitive-level coverage, and difficulty distribution of any candidate item bank must be demonstrably aligned to the UTME blueprint to protect fairness, score validity and defensible score interpretation.

Item Generation- LLM-based AIG Approach

This study employs an LLM- AIG approach to develop high-quality physics assessment items that align with the UTME blueprint. The study follows the general procedures described in our preliminary work (Omopekunola & Kardanova, 2024) for LLM-AIG with modifications. The following outline summarizes the item generation procedures with the modifications:

- 1) Developed a test blueprint grounded in the UTME Physics syllabus and analysis of past examination papers (2018–2023) both retrieved from the official JAMB portal (<https://www.jamb.gov.ng>) and accessible freely.

Bloom's taxonomy cognitive levels were operationalized using UTME learning objectives contained in the syllabus.

- 2) Utilized Gemini 2.0 flash LLM with instructional prompting (temperature was set to 0.0 and other parameters to default) to generate pool of multiple-choice questions aligned with the cognitive levels specified in Bloom's taxonomy. We extended the procedures from our earlier work by integrating novel elements domain and content specifications, in the generation pipeline designed to enhance item quality. The prompts clearly outlined the desired content domains (Mechanics, Light and Optics, Electricity, and Modern Physics) and specific topics within those domains. These are the broad domains in the Nigerian physics high school curriculum and UTME syllabus. The study focused exclusively on generating multiple-choice questions, because UTME has exclusively employed MCQs as its sole item format. Our generation architecture of LLM-based AIG is shown in figure 1.

For this study, 90 items were generated across all content domains and cognitive levels specified in the test blueprint (there were 31 blueprint cells). In our generation procedure we targeted at least two candidates on each blueprint cell while some received more than two. This aligns with best practices in assessment design which involves generating extra items to ensure a final form that meets content-coverage specifications. Sample of the generated item is shown in figure 2.

- 3) Evaluation of Generated Items.

The evaluation process involved SMEs (subject matter experts) who provided ratings for the generated test items. There were 3 SMEs from Nigeria. Selection criteria included possessing a master's degree in physics, at least seven years of experience teaching physics, and having prepared Senior Secondary School 3 (SS3) students (grade 12) for at least three exam cycles. If an expert failed to meet the first criterion, we accepted an alternative qualification of 12 years of teaching experience. The evaluation procedures

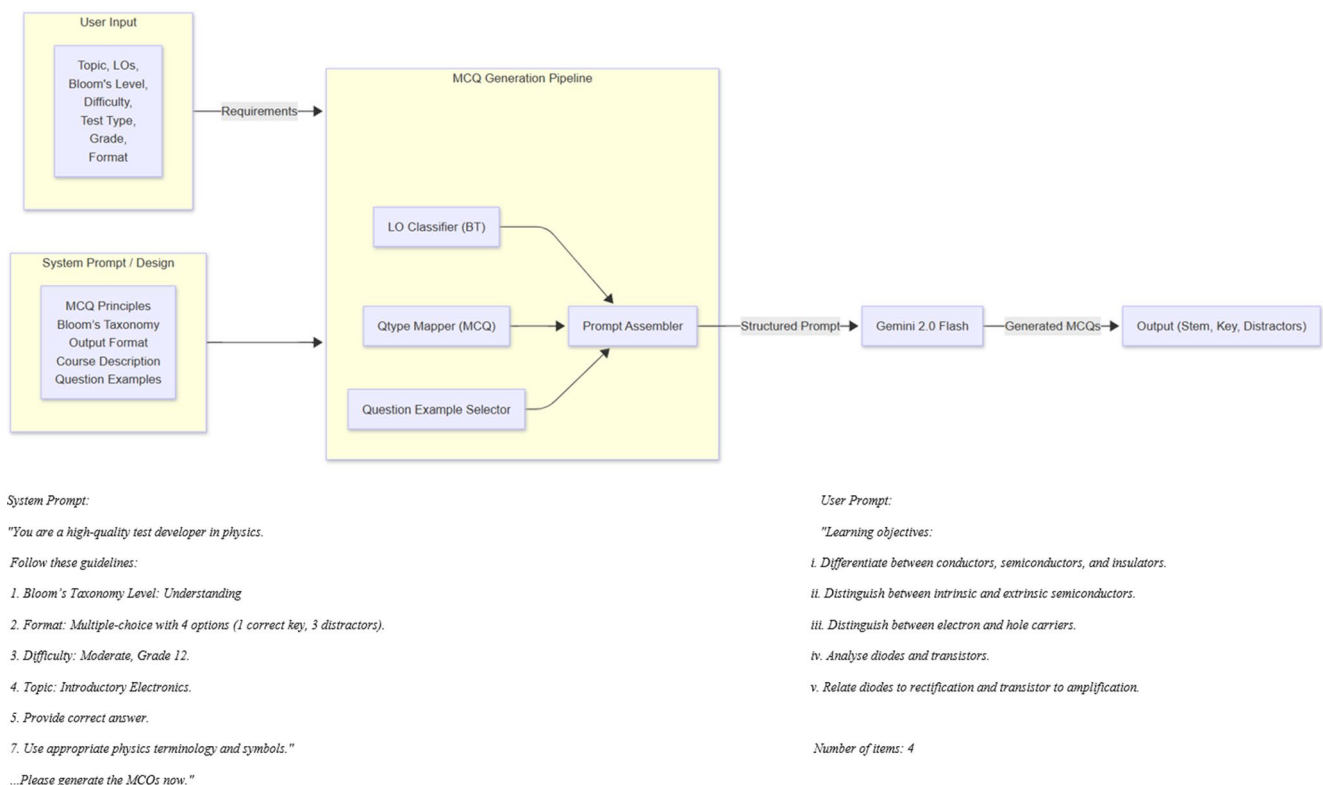


Fig. 1 Architecture of the automated MCQ generation pipeline with gemini 2.0 flash using instructional prompting

Fig. 2 Sample of item generated

Which of the following statements **BEST** describes the distinction between conductors, semiconductors, and insulators in terms of their band gaps?

- A. Conductors have a large band gap, semiconductors have a small band gap, and insulators have no band gap.
- B. Conductors have no band gap, semiconductors have a small band gap, and insulators have a large band gap.
- C. Conductors have a small band gap, semiconductors have no band gap, and insulators have a large band gap.
- D. Conductors and semiconductors have no band gap, while insulators have a large band gap.

integrate the evaluation frameworks of Doughty et al. (2024) and Chan et al. (2024) and Omopekunola and Kar-danova (2024), making a six comprehensive criteria: clarity, correctness, non-misleading content, use of appropriate language, alignment with the specified level of Bloom's taxonomy, and solvability with a single solution. Each criterion was scored on a binary scale, with 1 indicating a pass and 0 indicating a fail.

Piloting Phase

Instrument

Following the evaluation of items, the 3 SMEs reached a consensus on items that are qualified for the final test. Thus, 40 items were finally selected as final piloting instrument. This paired-generation strategy proved effective in practice in many blueprint cells at least one item passed SME review. The final 40-item form was chosen deliberately to mirror the operational UTME length, thus, 40 items were finally selected as final piloting instrument.

Sample

The piloting phase involved 527 SSS 3 (Grade 12) Nigerian students, aged 15 to 18 with a mean of 16.5 years, who were in their final year of high school. In Nigeria, class placement is based on students' cognitive ability and academic performance rather than age, hence the age spread. Also, Nigerian schools both provide and public operate a uniform curriculum, hence it is assumed that the test takers have a relatively the same prior knowledge. Topics covered are specified in the UTME physics syllabus which is reflected in the test blueprint (see Appendix 2 in supplementary file). The gender distribution was 43% girls and 57% boys. All participants had physics as their major subject and were selected from 12 schools (both private and government) across 9 cities in Nigeria, including the capital city.

Procedure

The test was administered in paper and pencil format by subject teachers under the same atmosphere and conditions as the actual UTME exam, including timing and other instructions. Research assistants supported the subject teachers when necessary. The test was administered in November, five months earlier than the real UTME test. After the test, materials were collected from the teachers for analysis. It is important to note that the tests were not conducted on the same day across all schools due to varying schedules set by school administrators; however, they were administered within a two-week interval, which is like the UTME. All

data were collected anonymously, with only serial numbers assigned to student booklets.

Psychometric Analysis

Psychometric analysis plays a crucial role in high-stakes examination creation to provide reliable and valid measurements. That is why we paid a lot of attention to the psychometric quality justification of the test.

The raw data was initially analyzed under Classical Test Theory (CTT) using the CTT package (Sheng, 2019) in R. Two item characteristics are considered in CTT—item difficulty and item discrimination (Crocker, 1986). All items were scored dichotomously (with correct=1 and incorrect=0), so an item difficulty is the proportion of examinees who answered it correctly. An item's discrimination shows how effectively it discriminates between high- and low-ability students (Crocker & Algina, 1986). The point biserial correlation was used as an index of item discrimination. It represents the Pearson correlation between an item score and total score and shows how closely performance on a test item is related to performance on the total test. Values of item discrimination may range from -1.00 to 1.00 , where values greater than 0.2 are usually considered acceptable.

Additionally, we conducted distractor analysis to evaluate the effectiveness of the incorrect response options (distractors) in multiple-choice items of the test and to be sure that each item contributes meaningfully to the assessment of the test-takers' knowledge or skills. Distractor analysis was performed in two steps. First, frequencies with which each distractor was chosen were analyzed. The purpose of this step is to identify ineffective distractors, which are not or rarely chosen by examinees (e.g. too obviously incorrect or not plausible). According to Abdulghani et al. (2014), a response frequency $< 5\%$ signals a poor distractor. Second, discrimination index for each distractor was calculated. Ideally, distractors should be chosen more frequently by low-performing test-takers, so their discrimination index should be negative.

Then, based on the findings from CTT analysis, item and test characteristics were evaluated under Item Response Theory (IRT) (Ayala, 2013). To identify which IRT model was the most appropriate for the data, we compared 1PL and 2PL models using the Akaike information criterion (Akaike, 1974) and Bayesian information criterion (Schwarz, 1978). The 1PL (One-Parameter Logistic) IRT model, also known as the Rasch model, assumes that all items have equal discrimination and differ only in difficulty, while the 2PL (Two-Parameter Logistic) IRT model allows both item difficulty and discrimination to vary across items. Lower AIC and BIC values indicate better fit. Then, we continued the analysis under the 2-parameter logistic model (2PL) using the TAM package in R (Robitzsch et al., 2020).

The analysis was performed in several steps as follows:

Step 1. Analysis of model fit: To measure the extent to which the data fit the model, we used the S-X² item-fit statistic (Orlando & Thissen, 2000). It evaluates whether an item's observed response behavior is consistent with the model-predicted probabilities, conditional on examinee ability. The statistic is approximately χ^2 -distributed, and small p-value (e.g., < 0.05) is considered for a potential misfit.

Step 2. Dimensionality study: Dimensionality was evaluated by fitting competing one-, two- and three-factor exploratory structural equation models (ESEM; Asparouhov & Muthén, 2009). Models were compared for fit and parsimony using information criteria (AIC, BIC) (Vrieze, 2012). After selecting the best-performing ESEM, local dependence and residual misfit were assessed with Yen's Q3 residuals (Yen, 1984). Values less than 0.2 are considered acceptable. These combined diagnostics were used to determine whether a unidimensional representation of the item set was adequate.

Step 3. Item analysis: At this stage, we examined the psychometric characteristics of individual test items.

Step 4. Reliability analysis: To analyze the reliability, we used the person reliability estimated using the EAP (Expected a Posteriori, Embretson & Reise, 2000).

Results

UTME Test Item Distribution and Characteristics

The UTME item distribution (see Appendix 1 in Supplementary file) shows items allocation by content domain, cognitive level, and difficulty. In this assessment, items were categorized into five cognitive dimensions: Remembering, Understanding, Applying, Analyzing, and Evaluating. There were 7 items (17.5%) on Remembering, 10 items (25%) on Understanding, 20 items (50%) on Applying, 1 item (2.5%) on Analyzing, and 5 items (5%) on Evaluating. The test items span core content areas such as Mechanics with 9 items (22.5%), 6 items (15%) on Heat and Thermodynamics, 6 items (15%) on Waves, 12 items (30%) on Electromagnetism, and 7 items (17.5%) on Modern Physics.

LLM-Generated Item Distribution and Expert Evaluation

The results of the item evaluation by experts demonstrated exceptionally high inter-rater reliability, with coefficients exceeding 0.90 across most of the criteria. Specifically, percentage agreement scores were 0.96 for clarity, 0.93 for

correctness, 0.98 for non-misleading content, 0.71 for non-misleading, 0.98 for alignment with Bloom's taxonomy, and a perfect score for appropriate language.

Following this evaluation Appendix 2 (in the Supplementary file) shows the test blueprint of 40 items selected for the test which also details the distribution of these items across both content domain and cognitive levels with difficulty levels. The cognitive breakdown is as follows: 2 items (5%) on Remembering, 11(27.5%) items on Understanding, 21(52.5) items on Applying, 2 items (5%) on Analyzing, and 5 items (12.5%) on Evaluating. This distribution was intentionally designed to mirror the structure of the real UTME Physics test, particularly emphasizing the applying level, which aligns with existing findings that problem-solving and procedural application dominate physics assessments (Motlhabane, 2017). Items were also distributed along the content areas, such as Mechanics with 12 items (30%), 6 items (15%) on Heat and Thermodynamics, 3 items (7.5%) on Waves, 10 items (25%) on Electromagnetism, and 9 items (22.5%) on Modern Physics.

Comparing the 2018 UTME blueprint and LLM blueprint, the LLM blueprint reflects the content distribution of the 2018 UTME in several core content areas. The blueprint features deliberate modifications, such as a 30% allocation of items in mechanics compared to 22.5% in the 2018 UTME, to accommodate expanded subtopic coverage based on the current UTME Physics syllabus and insights drawn from past examination papers (2018–2023). It was a deliberate decision not to replicate the 2018 UTME, rather created a blueprint using the current UTME physics syllabus and insight from the available past examination papers (2018–2023). These modifications were implemented to ensure that the blueprint accurately represents the test contents, which deviate a little from the 2018 UTME blueprint.

Psychometric Analysis

The analysis under the framework of the Classical Test Theory revealed that five items out of 40 had poor psychometric quality (low or negative discrimination, high difficulty, ineffective distractors). Review of these items showed that their topics are taught late in the Nigerian Physics curriculum, and they had not been yet covered at the time of testing. So, we assumed that guessing took place when students completed these tasks, and their poor performance in our study is explained by the fact that the testing was conducted several months earlier than the usual time for exam UTME and our testing also was not real high-stake exam, so the students were less motivated. Taking this into account, we decided to delete these 5 items from the further analysis.

We checked that deleting those 5 items does not affect the blueprint and the content balance, so we can continue

further with the test of 35 items. All these items had good psychometric quality under CTT. Difficulty of items varied in the range from 0.16 to 0.69 with the mean of 0.37. Moreover, discrimination and distractor analyses showed that all items had acceptable values of discrimination and distractors of all items functioned satisfactorily. Figure in Appendix 3 (in the Supplementary file) shows an example of a distractor plot.

Then we continued with IRT analysis (please see Appendix 3 (in the Supplementary) for details of analysis). Comparison between 1PL and 2PL models indicated a better model fit for 2PL model, so the 2PL model was selected for further analysis. Majority of items demonstrated perfect fit, while a few items exhibited a potential misfit ($0.1 < p < .05$); inspection of item characteristic curves for those items showed slight deviation of observed responses from the model predictions at high or low ability levels.

The analysis of the test dimensionality showed that the test is essentially unidimensional. The 2PL reliability is 0.79 and classical reliability $\alpha = 0.76$ for the test. Moreover, all items have good psychometric properties. Item difficulty varies widely, with a mean 1.16 and $SD = 1.35$ showing that the test covers a broad range of the latent ability continuum. Item discrimination values ranged also widely, with a mean 0.69 and $SD = 0.39$, indicating that most items demonstrated acceptable discrimination power. (See the Table 4 in Appendix 3 (in the Supplementary file) for details).

To conclude, psychometric analysis confirms the high quality of the test: it is essentially unidimensional, has high reliability, has good psychometric properties for all items, and can be used for a student's estimation.

Comparison of LLM and UTME Items

This section presents a comparative analysis of the difficulty distributions between LLM-generated physics items and those from the 2018 UTME. To validate these difficulty distributions through empirical psychometric analysis, a comparison was made between the LLM-generated physics items and the 2018 UTME item difficulty (Ibrahim, 2025), as shown in Table 1. We used the following classification of item difficulty (Rao et al., 2016): items with a classical difficulty value < 0.30 are considered difficult, those with values between 0.30 and 0.70 are considered ideal moderately difficult, and items with values > 0.70 are classified as easy. This approach provides an objective framework for evaluating the LLM-generated items and facilitated a direct

comparison with the difficulty distribution classically as observed in the 2018 UTME test.

The primary focus was not on numerical precision but on determining the proportion within the difficulty range. The result shows that the mean difficulty for LLM items (0.35) was lower than that for UTME items (0.53) (Ibrahim, 2025), with a close standard deviation. This implies that the LLM-generated items not only adhere to the aligned test blueprint, but also closely mirror the sample of the actual UTME test. However, LLM items are harder in general compared to the UTME items. Nonetheless, the robust expert evaluation was consistent with psychometric properties which provide strong evidence for the viability of LLM-based AIG.

Discussion of Results

Alignment of LLM-Generated Physics Items with the UTME Test Blueprint

The LLM test blueprint was designed to generate a test item set that aligns closely with the UTME blueprint for high-stakes physics examinations. Instead of selecting items arbitrarily, we systematically generated and combined items to ensure comprehensive coverage across all core topics and cognitive levels specified in the UTME framework. The analysis confirms that LLM demonstrated its capacity to produce items covering all domains and content areas. This targeted process ensured that the resulting item set fully reflects the real UTME test design in content and cognitive structure. By integrating items across these diverse domains and cognitive levels, we achieved a balanced and coherent assessment tool that aligns with the intended exam blueprint. The successful generation of items in every level not only validates our methodological approach but also demonstrates the potential of LLM-based automated item generation to develop high-quality items for high-stakes exams.

Comparison of LLM and UTME Item Distributions

The LLM test and the UTME test follow a similar pattern of item distribution, with the 2018 UTME test showing a lower mean score. This is due to the timing of the piloting phase, when some topics were not yet taught in schools, and the five-month gap between the piloting period and the actual UTME schedule.

Student motivation also plays a role in the difference in item difficulty between the LLM and UTME 2018 tests. High-stakes examinations like the UTME typically require heightened engagement and effort, while piloting exercises often lack extrinsic incentives, and students may not be equally motivated to perform at their highest ability. Some

Table 1 Comparison of items difficulties of LLM items and 2018 UTME

Test Type	Easy	Moderate	Difficult	Mean	SD
LLM Items	1 (2.9%)	28 (80%)	6 (17.1%)	0.37	0.11
2018 UTME	2 (5%)	33 (82.5%)	5 (12.5%)	0.53	0.13

UTME topics extend beyond the standard secondary school curriculum, sometimes necessitating an extra year of preparation for average ability students. Overall, these factors contribute to the difference in item difficulty between the LLM and UTME 2018 tests.

Psychometric Quality of LLM-Generated Physics Items for High-Stakes Assessments

This study demonstrates that LLM-generated physics items possess the necessary psychometric properties for high-stakes examinations like the Unified Tertiary Matriculation Examination (UTME). A two-framework approach integrating Classical Test Theory (CTT) and Item Response Theory (IRT) was used to assess item psychometric properties such as item difficulty, discrimination, and distractor effectiveness. Five items out of 40 were removed due to poor psychometric quality. However, the items skewed toward moderate–hard items for this sample. Two specific factors explain this result. First, the pilot was administered roughly five months earlier than the usual UTME cycle, and several items covered topics that schools typically teach later in the academic year; as a result, some students had not yet been taught all the content in the syllabus. Second, the pilot was administered under classroom conditions that do not fully reproduce UTME incentives; lower motivation and reduced test effort can depress observed accuracy. The remaining 35 items demonstrated good psychometrics quality.

Item Response Theory (IRT) analysis confirmed that these items are suitable for high-stakes testing. The most difficult item is from electric charges, aligning with Onuya et al. (2024), who reported that about 68% of students find electric charges to be the most challenging high school physics topic.

These findings demonstrate the potential of LLM-based AIG in supporting high-stakes assessments. Model comparisons revealed that 2PL model fits better than a 1PL model, indicating that both item difficulty and discrimination play important roles in estimating student abilities. This evidence demonstrates the potential of leveraging LLMs for high-stakes physics assessments, promising a way to expand item pools while maintaining psychometric qualities.

Reliability of LLMs in Generating High-Quality Physics Items for UTME Examinations

The reliability of the test for the University Test of Merit (UTME) is crucial for students to determine their eligibility for competitive STEM programs. The psychometric quality of LLM-generated physics items has been evaluated by experts, and psychometric analyses confirmed their validity

and reliability. The test items exhibit a considerable level of reliability, with a reliability coefficient of 0.79.

The results have implications for UTME and university admissions in Nigeria. Reliable LLM-generated physics items can standardize assessments across different exam cycles, minimizing discrepancies from human error or subjectivity in item creation. This consistency enhances fairness and transparency in the admissions process, ensuring all candidates are evaluated against the same psychometric standards. Integrating LLMs into item generation can significantly increase the scalability and efficiency of the UTME assessment process. Automated item generation offers a solution to maintain item quality while expanding the test item pool to prevent overexposure of certain items. Person reliability is sufficient for ensuring consistent measurements across the test population, suggesting that LLM-generated test items are suitable for high-stakes testing, where the ability to differentiate between students is paramount.

The findings have significant implications for AIG in physics. The study demonstrates that, with carefully designed prompts and expert vetting, state-of-the-art LLMs can generate physics items of considerable good psychometric quality. The adoption of LLMs for AIG in physics can substantially reduce the manual effort required for item development, allowing educators to focus more on instruction and student support. Integrating LLM-based Automated Item Generation (AIG) into national testing frameworks could improve fairness, efficiency, and consistency in university admissions, ensuring only the most capable candidates enter Nigeria's competitive higher education system. While this study focused on physics, our LLM-AIG can be applied to other STEM disciplines (such as chemistry, biology, computer science) that rely on procedural problem-solving, provided domain-specific prompt adjustments and empirical checks are implemented.

However, the study had some main limitations that should be carefully considered when interpreting and generalizing the findings. First, the findings are context-specific, as they are confined to physics items aligned with the Nigerian pre-university curriculum and the UTME blueprint. Generalizing these results to other subjects (such as biology, computer science) or educational systems (e.g., university-level assessments) requires further validation, as LLM performance may vary significantly across disciplines due to differences in concept, cognitive demands, and domain-specific knowledge. Second, the study used Gemini 2.0 flash for item generation, but the rapid evolution of LLMs means that newer models (Claude 3.5) or fine-tuned models specifically optimized for Physics or STEM tasks could yield varying results. This highlights the need for ongoing research to keep pace with advancements in LLMs capabilities. Third, we understand that Bloom's taxonomy used in the study can be not the best choice for physics, because Bloom's taxonomy assumes a progression from "lower-order"

skills to “higher-order” ones, but in physics (and in general in STEM) learning is often non-linear. But we decided to use Bloom’s taxonomy for the study mainly because UTME uses Bloom’s taxonomy for the operationalization of learning objectives and also because Bloom’s taxonomy is the most famous and popular among educators. Furthermore, the study focused primarily on multiple-choice items due to their prevalence in the UTME, leaving unexplored other assessment formats (such as open ended short and long) that require more complex structure. Future research should investigate LLMs’ capabilities in generating these more complex item types to provide a more comprehensive understanding of their usage in high-stakes assessments. Lastly, for future empirical studies, we recommend that following steps toward ensuring full content and cognitive coverage for the test blueprint before deployment: (1) re-balance the item bank to include items of sparse ability level (2) re-generate items explicitly for such ability where required; (3) re-pilot and perform IRT calibration nearer to the actual UTME administration so curriculum coverage. Also, there is growing need to design a multi-modal LLM-based framework that incorporates visual and open-ended items which are important components of physics assessments.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s10956-025-10289-y>.

Author Contributions All authors contributed to the study conception and design. The first draft of the manuscript was written by Moses Oluoke Omopekunola. Material preparation, data collection was performed by Moses Oluoke Omopekunola. Formal analysis, review and editing, funding acquisition, supervision and resources were performed by Elena Yu Kardanova. All authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

Funding This research was sponsored by the grant of Russian Scientific Fund, with reference number 25-18-00751.

Data Availability The items and other relevant data (with the analysis code) used in this study are available in the supplementary file. Supplementary data to this article can be found online at <https://github.com/MOO-DIO/LLM-paper.git>.

Declarations

Ethical Approval Approval was gotten from the school heads, following conventional research ethics in the country.

Consent to Participate Consent for participation was obtained from all schools.

Informed Consent Not Applicable.

Consent to Publish Not Applicable.

Statement Regarding Research Involving Human Participants and/or Animals Not Applicable.

Competing interests The authors declare no conflicts of interest.

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